Two-body interactions from the tensor-product spectral triple: Selection rules from the algebra, coupling strengths from the metric

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We construct the tensor-product spectral triple $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^3}$ for two particles on the Fock-projected three-sphere at finite cutoff $n_{\max}$, and investigate whether the gauged spectral action generates the Coulomb inter-particle interaction from the framework's own algebraic structure. The free tensor-product Dirac $D_{\text{total}} = D_1 \otimes I + \gamma_1 \otimes D_2$ satisfies $\{D, \gamma\} = 0$ exactly, so D^2_{total} factorizes and the free spectral action contains no interaction (Theorem 1). Gauging via the inner fluctuation $A = \sum_i a_i^\dagger [D, a_i]$ from the truncated operator system $\mathcal{O}_{n_{\max}}$ produces a genuine two-body interaction: the connected (non-factorizable) fraction of $\{D, A\}$ is 77% at $n_{\max} = 2$ and 32% at $n_{\max} = 3$ (Proposition 3). This interaction is supported only on the direct (density–density) elements $\langle ab|\cdot|ab\rangle$, so it is m-conserving, $\Delta l = 0$ and pure $k = 0$ by construction (Theorem 4); the exchange and $k > 0$ channels of $1/r_{12}$ are absent. The radial matrix element ratios do not match the Coulomb $1/r_{12}$ kernel: the coefficient of variation exceeds 1600% and the Pearson correlation is 0.41–0.58 (Proposition 6). These results precisely delineate Paper 31's universal/Coulomb partition at the two-body level: the gauged *algebra* yields a monopole, density–density subset of the Coulomb channels; the *metric* determines the coupling strengths. The multi-focal composition wall identified across six independent observables in the GeoVac programme is structurally located at this algebra/metric boundary.

I. INTRODUCTION

The GeoVac framework constructs a discrete spectral triple $(\mathcal{A}, \mathcal{H}, D)$ on the Fock-projected three-sphere [1, 8]: the algebra $\mathcal{A}$ of truncated multipliers on S^3, the Hilbert space $\mathcal{H}$ of Fock-projected states, and the Camporesi–Higuchi Dirac operator D. Truncations of the true Camporesi–Higuchi window converge to the continuum S^3 in the van Suijlekom state-space Gromov–Hausdorff distance [10] (the chirality-doubled model used below is not the object of that theorem), and the spectral action $\mathrm{Tr}\, f(D^2/\Lambda^2)$ reproduces the standard continuum Chamseddine–Connes behaviour [11].

A central open question is whether the framework can derive *inter-particle interactions* from its own structure, rather than importing them as external input. The electron–electron Coulomb repulsion $V_{ee} = 1/r_{12}$, the nucleon–nucleon force, and the core–valence interaction in molecular systems are all two-body couplings that the framework currently imports from standard quantum mechanics. The "multi-focal composition wall" [9] — the observation that the framework couples discrete quantum numbers cleanly but has no native composition theorem for spatial focal lengths — has been documented across six independent observables [16].

In this paper we attack the problem directly. We construct the tensor-product spectral triple $\mathcal{T}_1 \otimes \mathcal{T}_2$ for two particles, each on a copy of the truncated S^3, and compute the gauged spectral action to determine what interaction, if any, emerges from the algebra's inner fluctuations.

The results are sharp: the gauged interaction is consistent with the angular selection rules but reproduces only their monopole, density–density subset (m-conserving, $\Delta l = 0$, $k = 0$), and its radial coupling strengths are not proportional to the Coulomb kernel. This places the multi-focal wall precisely at the boundary between the algebra $\mathcal{A}$ (which determines angular structure) and the Dirac operator D (which determines radial couplings) — the universal/Coulomb partition of Paper 31 [7], now verified at the two-body level.

II. SETUP

A. Single-particle spectral triple

We work with the scalar sector of the Fock-projected S^3 at cutoff $n_{\max}$. The Hilbert space is

$$\mathcal{H}_{n_{\max}} = \mathrm{span}\{|n, l, m\rangle : 1 \le n \le n_{\max}\}, \tag{1}$$

with $\dim \mathcal{H}_{n_{\max}} = N = \sum_{n=1}^{n_{\max}} n^2$. For the Dirac structure, we chirality-double to $\hat{\mathcal{H}} = \mathcal{H}^+ \oplus \mathcal{H}^-$ with $\dim \hat{\mathcal{H}} = 2N$.

The Dirac operator is the Camporesi–Higuchi form [12]:

$$D|n, l, m, \chi\rangle = \chi \left(n + \tfrac{1}{2}\right) |n, l, m, \chi\rangle, \tag{2}$$

where $\chi = \pm 1$ is the chirality. The chirality operator γ swaps the two sectors: $\gamma|n, l, m, +\rangle = |n, l, m, -\rangle$.

The algebra $\mathcal{A} = \mathcal{O}_{n_{\max}}$ is the Connes–van Suijlekom [14] truncated operator system, generated by the 3-Y multiplier matrices M_{NLM} with entries

$$(M_{NLM})_{(nlm),(n'l'm')} = \int_{S^3} \overline{Y^{(3)}_{nlm}} \cdot Y^{(3)}_{NLM} \cdot Y^{(3)}_{n'l'm'}\, d\Omega. \tag{3}$$

These are embedded into $\hat{\mathcal{H}}$ as $\hat{M} = \mathrm{diag}(M, M)$, acting identically on both chirality sectors.

B. Tensor-product Dirac

For two particles, the total Hilbert space is $\hat{\mathcal{H}}_1 \otimes \hat{\mathcal{H}}_2$ with dimension $(2N)^2$. The tensor-product Dirac operator is the standard product [13]:

$$D_{\text{total}} = D_1 \otimes I_2 + \gamma_1 \otimes D_2. \tag{4}$$

The chirality twist γ_1 ensures D_{total} is a first-order operator on the graded tensor product.

III. FREE FACTORIZATION

Theorem 1 (Free factorization)**.** *For the Camporesi–Higuchi Dirac* (2) *on the chirality-doubled scalar basis,* $\{D, \gamma\} = 0$ *exactly. As a consequence,*

$$D_{\text{total}}^2 = D_1^2 \otimes I_2 + I_1 \otimes D_2^2, \tag{5}$$

so the heat trace factorizes exactly, $\operatorname{Tr} e^{-tD_{\text{total}}^2} = \operatorname{Tr} e^{-tD_1^2} \cdot \operatorname{Tr} e^{-tD_2^2}$ *for every* $t > 0$*; for a general cutoff function* f *the spectral action is a sum of* f *over sums of single-particle eigenvalues, with no cross term.*

Proof. The Dirac eigenvalue on $|n, l, m, \chi\rangle$ is $\chi(n + \frac{1}{2})$. The chirality operator sends $\chi \to -\chi$. For any basis state,

$$(D\gamma + \gamma D)|n, l, m, \chi\rangle = \chi(n+\tfrac{1}{2})(-\chi)(n+\tfrac{1}{2}) + (-\chi)(n+\tfrac{1}{2})\chi(n+\tfrac{1}{2}) = 0,$$

using $D|n, l, m, -\chi\rangle = -\chi(n + \frac{1}{2})|n, l, m, -\chi\rangle$. Hence $\{D, \gamma\} = 0$. Expanding D_{total}^2:

$$D_{\text{total}}^2 = D_1^2 \otimes I + (D_1\gamma_1 + \gamma_1 D_1) \otimes D_2 + I \otimes D_2^2.$$

The cross term vanishes by $\{D_1, \gamma_1\} = 0$. □

Remark 2. *This is the correct physical statement: free particles do not interact. The spectral action of the free Dirac gives independent single-particle kinetic energies. Any interaction must enter through* gauging*.*

Numerical verification. At $n_{\max} \in \{2, 3\}$, the heat trace ratio $\operatorname{Tr}(e^{-tD_{\text{total}}^2})/[\operatorname{Tr}(e^{-tD_1^2}) \cdot \operatorname{Tr}(e^{-tD_2^2})] = 1.000000$ to machine precision at all tested $t \in \{0.1, 0.5, 1.0, 2.0\}$.

IV. GAUGED INTERACTION

The inner fluctuation of D_{total} by the tensor-product algebra $\mathcal{A}_1 \otimes \mathcal{A}_2$ is

$$[D_{\text{total}}, \hat{M}_i \otimes \hat{M}_j] = [D_1, \hat{M}_i] \otimes \hat{M}_j + \gamma_1 \hat{M}_i \otimes [D_2, \hat{M}_j]. \tag{6}$$

Both terms are generically nonzero: $[D, \hat{M}]$ is nonzero whenever $\hat{M}$ has cross-shell matrix elements (coupling states with different n, hence different D-eigenvalues). At $n_{\max} = 2$, four of 14 multiplier generators have nonzero commutators; at $n_{\max} = 3$, 29 of 55.

TABLE I. Heat trace ratio $\operatorname{Tr} e^{-t(D+\varepsilon A)^2}/[\operatorname{Tr} e^{-tD_1^2} \cdot \operatorname{Tr} e^{-tD_2^2}]$ at $t = 0.5$ for the full-algebra gauge field.

ε	$n_{\max} = 2$	$n_{\max} = 3$
0.0	1.000000	1.000000
0.1	0.995465	0.838438
0.5	0.977815	0.410165
1.0	0.956832	0.185405

The *rotationally invariant* one-form is constructed using the conjugate pairing [13]:

$$A = \sum_{(NLM)} \sum_{(N'L'M')} (\hat{M}^\dagger_{NLM} \otimes \hat{M}^\dagger_{N'L'M'}) \cdot [D_{\text{total}}, \hat{M}_{NLM} \otimes \hat{M}_{N'L'M'}], \tag{7}$$

summing *independently* over all allowed multiplier labels on each tensor factor in $\mathcal{O}_{n_{\max}}$. This is the full inner fluctuation of the product algebra $\mathcal{A} \otimes \mathcal{A}$, not only the diagonal ($N'L'M' = NLM$) terms; all numerical results in this paper use this double sum. The dagger pairing ensures m-conservation (total magnetic quantum number $M_1 + M_2$ is preserved) and Hermiticity of A.

Proposition 3 (Connected interaction)**.** *The anticommutator* $\{D_{\text{total}}, A\}$ *contains a non-factorizable (connected) two-body component. Define the connected part via partial-trace subtraction:*

$$V_{\text{conn}} = \{D, A\} - \frac{1}{d_2}\operatorname{Tr}_2\{D, A\} \otimes I_2 - I_1 \otimes \frac{1}{d_1}\operatorname{Tr}_1\{D, A\} + \frac{\operatorname{Tr}\{D, A\}}{d_1 d_2} I_1 \otimes$$

At $n_{\max} = 2$*:* $\|V_{\text{conn}}\|/\|\{D, A\}\| = 76.7\%$*. At* $n_{\max} = 3$*:* $\|V_{\text{conn}}\|/\|\{D, A\}\| = 32.4\%$*.*

The interaction is genuine: the connected fraction is nonzero, and the heat trace $\operatorname{Tr} e^{-t(D+\varepsilon A)^2}$ departs from the factorized value monotonically with ε (Table I).

V. ANGULAR SELECTION RULES

Theorem 4 (Angular structure of the connected interaction)**.** *The connected two-body interaction* V_{conn} *from the rotationally invariant gauge field* (7) *is supported only on the direct elements* $V_{\text{conn}}[a, b, a, b]$ *at* $n_{\max} \in \{2, 3\}$ ***[MEASURED]****, so items (i)–(iii) hold for it automatically; they show that it is consistent with the Coulomb selection rules, not that it generates them. It satisfies:*

(i) ***m-conservation:*** $m_1 + m_2 = m_1' + m_2'$ *for all nonzero matrix elements — structural for all n (the* $M^\dagger \cdot [D, M]$ *conjugate pairing forces the gauge field's total* $M = 0$*), confirmed numerically at* $n_{\max} \in \{2, 3\}$*.*

(ii) ***Gaunt compatibility:*** $|\Delta l_1| = |\Delta l_2| = k$ *for all nonzero* $(l_1, l_2) \to (l_1', l_2')$ *channels, with* $k = 0$ *on the direct support (consistent with the potential-independent angular sparsity theorem of Paper 22 [6]), confirmed at* $n_{\max} \in \{2, 3\}$.

(iii) ***Pure monopole:*** 100% *of* $\|V\|^2$ *is in the* $k = 0$ *multipole channel,* verified at $n_{\max} \in \{2, 3\}$ *(computational; not asserted beyond the tested cutoffs).*

(iv) ***Hermiticity and exchange symmetry*** *hold to machine precision* ($< 10^{-17}$).

Verification. Direct computation from the 4-index tensor $V_{\text{conn}}[a, b, c, d]$ extracted from the positive-chirality sector at $n_{\max} = 2$ ($N = 5$, 25 nonzero elements) and $n_{\max} = 3$ ($N = 14$, 196 nonzero elements). The m-conservation is enforced by the $M^\dagger \cdot [D, M]$ conjugate pairing, which ensures the total M quantum number of the gauge field is zero. Every nonzero element is direct, $[a, b, a, b]$, so (ii) and (iii) hold with $\Delta l = 0$. All four items (m-conservation, Gaunt compatibility, 100% $k = 0$ monopole, and the connected-fraction proposition above) are regression-pinned in `tests/test_paper31_two_body.py` (`test_spectral_action_two_body_nmax2` and `..._nmax3_decreasing`), which also verifies the radial-vs-Coulomb Pearson decreasing with $n_{\max}$. □

Remark 5. *The* $k = 0$ *monopole channel corresponds to the* F^0 *Slater integral in the Neumann expansion of* $1/r_{12}$ *(Paper 12 [2]). The exchange and higher-multipole channels of* $1/r_{12}$*, already present at these cutoffs, are absent from* V_{conn}*, which is direct-diagonal.*

VI. RADIAL COMPARISON WITH COULOMB

Proposition 6 (Radial weights do not match Coulomb)**.** *The matrix element ratios* $V_{\text{conn}}[a, b, c, d]/\langle ab|r_{12}^{-1}|cd\rangle$ *are not constant across the nonzero elements. At* $n_{\max} = 2$*: coefficient of variation* CV = 16.5*, Pearson correlation* $r = 0.58$*, sign agreement* 68%*. At* $n_{\max} = 3$*:* CV = 20.3*,* $r = 0.41$*, sign agreement* 54%*.*

The spectral-action interaction is a strict *subset* of the Coulomb matrix elements: all nonzero spectral elements also appear in the Coulomb matrix (0 spectral-only entries), but the Coulomb matrix has many additional nonzero elements (82 at $n_{\max} = 2$, 3 604 at $n_{\max} = 3$, against 25 and 196 spectral ones) that are zero in the spectral interaction. The Pearson correlation *decreases* from $n_{\max} = 2$ to 3, consistent with a structural rather than a finite-size mismatch.

Remark 7. *The* $1/r_{12}$ *Coulomb kernel in the Fock-projected basis arises from the chordal distance identity under stereographic projection (Paper 7 [1]). This is a* metric *property of* S^3*, not a spectral-action property. The heat-kernel versus Green's-function contrast does not explain the radial mismatch, however: the resolvent route fails the same way (Sec. IX).*

VII. THE ALGEBRA/METRIC BOUNDARY

The results of Sections V–VI precisely delineate the universal/Coulomb partition of Paper 31 [7] at the two-body level:

- The **algebra** $\mathcal{A}$ (truncated operator system, 3-Y multipliers) determines the angular selection rules: Gaunt compatibility, m-conservation, and the multipole hierarchy. These are *universal* — they hold for any central potential, as proved by the angular sparsity theorem of Paper 22 [6].
- The **Dirac operator** D (Camporesi–Higuchi eigenvalues) determines the radial coupling strengths. The gauged spectral action generates an interaction whose radial weights depend on the specific spectrum of D, not on the $1/r$ Coulomb potential. The Coulomb kernel is additional input from the Fock projection's metric structure.

This decomposition explains the "multi-focal composition wall" documented across six observables in the GeoVac programme [9]: the wall sits at the $\mathcal{A}/D$ boundary. The framework couples discrete labels (angular quantum numbers) via the algebra's inner fluctuations, but the spatial coupling strength (radial integrals) requires the metric — the Fock projection that maps the graph onto S^3, and with it, the $1/r$ potential.

Corollary 8. *For two particles on* $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^3}$*, the inter-particle interaction splits as*

$$V_{12} = \underbrace{\textit{angular selection rules}}_{\textit{from } \mathcal{A}} \times \underbrace{\textit{radial coupling strengths}}_{\textit{from } D/\textit{metric}}. \tag{8}$$

The first factor is the angular algebra's (Paper 22 [6]); the inner fluctuation here populates only its monopole, direct subset and does not generate it. The second factor is calibration data (the Slater integrals R^k *of Paper 12 [2]).*

VIII. IMPLICATIONS

A. For atomic and molecular structure

The result vindicates the composed-geometry architecture of Paper 17 [4]: the Gaunt selection rules that give the exactly Q-linear composed Pauli count ($N_{\text{Pauli}} = 27.90 \times Q$ at fixed basis) of Paper 14 [3] are those of the angular algebra (Paper 22 [6]); the tensor-product inner fluctuation is consistent with them but reproduces only their monopole, direct subset. The radial Slater integrals that complete the Hamiltonian are the irreducible

calibration content, consistent with Paper 18's exchange-constant taxonomy [5] (calibration tier).

B. Connection to double factorization and tensor hypercontraction

The two-electron operator decomposition $V_{ee} = \sum_P L_P \otimes L_P$ used in double factorization [17] and tensor hypercontraction [18] is literally an element of $\mathcal{A} \otimes \mathcal{A}$ in the tensor-product spectral triple constructed here. In the GeoVac basis, this decomposition is supplied analytically by the multipole expansion

$$\frac{1}{r_{12}} = \sum_{L,M} \frac{4\pi}{2L+1} \frac{r_<^L}{r_>^{L+1}} Y_{LM}(\Omega_1)\, Y_{LM}^*(\Omega_2),$$

with L_P proportional to Y_{LM} and rank equal to the number of distinct radial-density channels $\rho_\alpha(r) = R_{n_a l_a}(r)\, R_{n_b l_b}(r)$ at each angular momentum L.

Numerical double factorization applied to a GeoVac composed ERI tensor reproduces this analytic decomposition bit-exactly at the production basis: the relative reconstruction error $\|G - VKV^\top\|_F / \|G\|_F$ between the empirical ERI tensor and its multipole-VKV form is 10^{-16} at $n_{\max} = 2$ and 10^{-14} at $n_{\max} = 3$, i.e. machine precision. Every singular vector of the reshaped two-electron tensor lives strictly inside the multipole-channel subspace at every basis size tested ($n_{\max} \in \{2, 3, 4\}$), with total subspace overlap unity to machine precision. At the production basis ($n_{\max} = 2$, valence $\{1s, 2s, 2p\}$), the double-factorization rank equals the analytic distinct-radial-density count exactly — seven, distributed as four densities at $L = 0$, two at $L = 1$, one at $L = 2$.

At larger basis ($n_{\max} \geq 3$), the radial-integral matrix $R^L(\rho_\alpha; \rho_\beta) = \langle \rho_\alpha | r_<^L / r_>^{L+1} | \rho_\beta \rangle$ develops a graded singular-value spectrum that decays over many orders of magnitude rather than terminating at a clean cutoff. This is the well-known Laguerre-basis ill-conditioning of hydrogenic ERIs — the same phenomenon that motivates resolution-of-identity and Cholesky methods in standard atomic and molecular structure. Radial densities with similar effective exponents $(1/n_a + 1/n_b)/2$ produce nearly proportional contributions against the Coulomb kernel, so the apparent "rank" of R^L depends on threshold. The additional compression that DF finds beyond the bare analytic multipole count is therefore not a new algebraic structure but a numerical realization of the same spectral content, exposed through SVD ordering. Operationally: double factorization on a GeoVac Hamiltonian is the multipole expansion accessed by SVD rather than by closed form. This is a numerical observation about how DF orders an already-established spectral content (the analytic multipole count, whose $k = 0$ monopole dominance is pinned in `tests/test_paper31_two_body.py`), not a separate algebraic claim.

C. For nuclear structure

The same $\mathcal{A}/D$ split applies to the nucleon–nucleon interaction. The Minnesota potential's angular structure (central force, spin-dependent channels) is algebra-side; the radial Gaussian form factors ($V_R e^{-\kappa_R r^2}$, etc.) are D-side calibration. This explains why the nuclear model requires imported potentials: the framework derives *which* nuclear channels couple but not the NN force strength.

D. For gravity

The gravity sector (Paper 51 [11]) is special: the spectral action directly gives metric properties (Einstein–Hilbert action, Newton's constant, cosmological constant) because gravity *is* the metric. The two-body Coulomb interaction lives on the other side of the $\mathcal{A}/D$ boundary, consistent with the gravity sector being reproduced at the continuum level (Paper 51) while inter-particle interactions require imported potentials.

IX. DISCUSSION

The tensor-product spectral action generates genuine two-body physics: a non-factorizable interaction consistent with the angular selection rules, populating only their monopole, direct subset. That the radial weights do not match Coulomb is a measured negative **[MEASURED]** (one gauge construction, $n_{\max} = 2, 3$). It is not explained by heat-kernel versus Green's-function structure: the resolvent route, which supplies a Green's function, fails the same way (Paper 31 [7], two-body section; best Pearson 0.85, falling with $n_{\max}$), and the mismatch is attributed to the Fock projection's conformal factor, a function rather than a constant.

The practical consequence is that inter-particle radial integrals (Slater R^k, Minnesota form factors, Clementi–Raimondi screening) are irreducible calibration data in the GeoVac framework. The angular selection rules that govern which integrals are nonzero are framework-native. This is a more precise statement of the multi-focal composition wall: it is not a missing tool but the structural boundary between what the algebra determines and what the metric determines in a spectral triple.

Open question. Whether the Green's function kernel can be recovered from the spectral triple by a construction other than the spectral action — for example via the Connes distance function — remains an open problem. The resolvent $(D^2)^{-1}$ has been tested and does not reproduce $1/r_{12}$ (four weightings, Pearson at most 0.85 and falling with $n_{\max}$; Paper 31 [7]).

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